

Effect of Major NBA Trades on Social Media Discourse: An Analysis of the 2026 NBA Trade Deadline

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Abstract

This paper examines a reply network constructed from Bluesky NBA posts collected during the week surrounding the 2026 NBA trade deadline to investigate how trade events influence the volume and structure of online discourse. In this directed, weighted network, nodes represent users posting or replying to NBA posts, edges represent replies between users, and edge weights denote the frequency of such replies. We find that the network is highly disconnected and exhibits a core-periphery-like topology with a power-law-like in-degree distribution, indicating strongly concentrated attention on a small number of users. Prominent hubs correspond primarily to NBA insiders and information brokers who report trades and NBA-related news in real time. A temporal analysis further reveals substantial spikes in posting activity immediately following the initial reporting of major trades. The largest surges in activity closely align with the first trade announcements of star players made by leading NBA insider Shams Charania on Twitter, suggesting that key information brokers play a central role in shaping discourse.

Introduction

The NBA trade deadline is one of the most consequential dates on the league calendar, marking the final day teams can trade players before the offseason. Players may be exchanged for other players, draft picks (future selection rights), cash considerations, or combinations thereof. Following the deadline, team rosters are largely finalized for the remainder of the regular season and the playoffs, prompting many organizations to make last-minute roster adjustments [9]. As a result, the deadline generates substantial league-wide speculation and discussion, with fans debating hypothetical trades and reacting to newly completed deals.

The 2026 NBA trade deadline, the subject of our analysis, occurred on February 5th at 3:00 PM EST. Notably, 23 of the 26 trades completed between the start of the season on October 21st and the deadline were between February 3rd and February 5th, the final two days before the deadline [1]. This clustering of trades is common, as teams often accelerate negotiations and finalize deals under deadline pressure.

Historically, much of the buzz from the deadline was talked about in person; however, social media has largely

supplanted these interactions. The primary platform for NBA trade discussion has been Twitter, now known as X, following its rebranding under Elon Musk. Many trades are first announced on X by prominent NBA insiders. Trade announcements became popularly known as “Woj Bombs,” referring to Adrian Wojnarowski, who frequently broke major trade news before his retirement in September 2024. The sudden surge of online reactions following his reports resembled an information “bomb,” with some posts receiving tens of thousands of likes and replies [7].

Since Wojnarowski’s retirement, several insiders have competed to occupy the leading role in NBA trade reporting. The most prominent has been ESPN’s Shams Charania, who has broken numerous major trade stories, including the 2025 trade that sent Luka Dončić to the Los Angeles Lakers—widely regarded as one of the most significant trades of the 21st century [8].

Although discourse on X remains influential, changes following Musk’s 2023 acquisition—including the restriction and monetization of the Twitter API—have made large-scale X data collection difficult for researchers. Since Musk’s takeover, many offshoots of Twitter have launched, including Bluesky, which was made public in 2024 as an alternative platform designed to recreate aspects of the early Twitter experience.

Bluesky now has over 40 million registered users and offers similar features, including posts, reposts, likes, and replies. The platform is decentralized, allowing users to select their own servers and domains, though many use the default .bsky.social domain. Users may share identical usernames across different domains [2]. Importantly for this study, Bluesky provides a more accessible API compared to X, enabling large-scale data collection [11].

This paper analyzes the flow of NBA-related information on Bluesky during the 2026 trade deadline week. Specifically, it addresses two research questions: (1) What is the structural organization of the Bluesky NBA reply network during the trade deadline, and does it exhibit centralized hub formation? (2) How does the structure of the reply network evolve in response to “Woj bombs”, aka breaking trades on star players?

Related Work

Previous research has examined how “exceptional events” generate rapid shifts in online discourse. Studies of major events such as the discovery of the Higgs boson in 2012 and the People’s Climate March in 2014 document substantial spikes in tweet volume immediately following the event. In these cases, reply networks exhibited very low clustering coefficients (approximately 0.03) and scale-free in-degree distributions across replies, retweets, and mentions, indicating highly unequal attention concentration. Additionally, users were found to employ replies, retweets, and mentions for distinct communicative purposes rather than interacting with identical content in the same way [10].

These findings suggest a broader pattern: real-world shocks can produce rapid surges in online discourse while simultaneously reinforcing centralized hubs, in which a small number of accounts receive a disproportionate share of interactions.

Similar dynamics have been observed during major sporting events. A study of the 2019 NBA Finals between the Golden State Warriors and the Toronto Raptors applied temporal and sentiment analysis to Twitter data and found that tweet volume decreased at night, increased near the end of games, following key incidents (e.g., Kevin Durant’s Achilles injury), and around noon the following day. Tweet density was also higher in geographic regions associated with the teams or key players. Sentiment fluctuated with game outcomes; winning teams had more positive tweets and losing teams had more negative tweets. Two main fan behaviors reoccurred: “BIRGing” (basking in reflected glory), in which fans publicly associate more strongly with their team when they win, and “blasting,” in which fans ridicule underperforming opponents [5].

Together, these works suggest that sports events can be modeled as “exceptional events” with significant moments leading to increased discourse.

Beyond Twitter, research on Reddit has demonstrated that online discussion networks often exhibit sparse connectivity, small-world properties, and power-law in-degree distributions. One study constructing 72 multilayer subreddit networks found that geographically based and sports-oriented subreddits displayed high levels of deliberation. In comment-to-comment networks, 78% of actors belonged to giant components (defined as users with at least three posts), and the networks exhibited higher global clustering coefficients than comparable random graphs, along with scale-free in-degree distributions [4].

Across platforms, these studies point to a consistent structural pattern: online discourse following significant events tends to organize around highly connected hubs, producing heavy-tailed in-degree distributions even while overall network density remains sparse.

However, two gaps remain. First, prior work has largely focused on general “exceptional events” or live sporting contests, rather than deadline-driven institutional events such as the NBA trade deadline. Unlike games, which unfold continuously, trade deadlines concentrate uncertainty into a fixed temporal window, potentially privileging accounts that first report breaking news. Second, while Twitter and Reddit net-

works have been extensively analyzed, comparatively little work has examined whether similar structural dynamics emerge on newer, decentralized platforms such as Bluesky.

This paper bridges these gaps by investigating whether NBA trade announcements function as event-driven attention shocks that produce rapid spikes in discourse volume and increased concentration of reply activity around key information brokers (NBA insiders). By analyzing the structural and temporal dynamics of the Bluesky NBA reply network during the 2026 trade deadline, we test whether previously observed scale-free and hub-dominated patterns generalize to a newer platform and to a deadline-based sporting event context.

Dataset and Data Processing

The Bluesky NBA posts dataset was collected using the Bluesky API [11]. Data were gathered from midnight on February 2nd to midnight on February 8th, 2026. This time window was chosen to capture NBA discourse surrounding the trade deadline, which occurred on February 5th at 3:00 PM EST, including several days both before and after the event. Due to the large volume of data and Bluesky’s API rate limits, collecting a substantially larger dataset was impractical given the scope of this study.

Posts on Bluesky are timestamped in Coordinated Universal Time (UTC). To ensure the dataset began at midnight Eastern Standard Time (EST), data collection started at 5:00 AM UTC and was later converted to EST.

Posts flagged as “NBA posts” were identified using a curated set of keywords. These keywords included terms such as “NBA” and “trade deadline,” as well as team names and team abbreviations. A complete list of these keywords is provided in Table 1.

Other potential keywords, such as “basketball” or “trade,” were considered but excluded because they produced a large number of unrelated posts. For example, the keyword “trade” frequently returned posts about financial markets, while “basketball” captured many posts about college basketball. Player names were also excluded because the large number of NBA players could bias the dataset toward discussions involving specific individuals. Due to the conservative set of keywords used, it is likely that some NBA-related posts were not captured, and future work could refine these filters to improve coverage.

For each NBA post identified, its entire reply tree was collected up to a depth of 100 replies. This depth was chosen so that it would effectively include the entire reply tree. Replies were included even if they did not contain any NBA-related keywords. The reply tree consists of all direct replies to the original post as well as replies to those replies. Self-loops—instances in which users replied to their own posts—were excluded from the analysis, as they did not represent natural NBA discourse and would skew future centralization metrics.

For each post, we collected the `User DID`, which is a unique identifier for the user, the `User Handle`, the `Post ID`, the `text` of the tweet, and the `date` of the tweet.

For each reply, the dataset includes the `Source DID` (identifier of the replying user), the `Source Handle`,

Table 1: Bluesky NBA Keywords

Category	Terms
League Terms	NBA, trade deadline
Team Names	Atlanta Hawks, Boston Celtics, Brooklyn Nets, Charlotte Hornets, Chicago Bulls, Cleveland Cavaliers, Dallas Mavericks, Denver Nuggets, Detroit Pistons, Golden State Warriors, Houston Rockets, Indiana Pacers, Los Angeles Clippers, Los Angeles Lakers, Memphis Grizzlies, Miami Heat, Milwaukee Bucks, Minnesota Timberwolves, New Orleans Pelicans, New York Knicks, Oklahoma City Thunder, Orlando Magic, Philadelphia 76ers, Phoenix Suns, Portland Trail Blazers, Sacramento Kings, San Antonio Spurs, Toronto Raptors, Utah Jazz, Washington Wizards
Abbreviations	#ATL, #BOS, #BKN, #CHA, #CHI, #CLE, #DAL, #DEN, #DET, #GSW, #HOU, #IND, #LAC, #LAL, #MEM, #MIA, #MIL, #MIN, #NOP, #NYK, #OKC, #ORL, #PHI, #PHX, #POR, #SAC, #SAS, #TOR, #UTA, #WAS

the Source Post ID, the Target DID (identifier of the user being replied to), the Target Handle, and the Parent Post ID (the post being replied to). Additionally, edge weights were calculated to represent the number of times one user replied to another.

In the resulting network, nodes represent users who posted or replied to NBA content. A directed edge from user A to user B indicates that user A replied to user B. Edge weights represent the number of times user A replied to user B.

In total, the dataset contains 36,349 unique NBA posts, 8,104 unique users posting or replying to NBA content (nodes), 8,617 reply interactions (edges), and 6,936 unique user-to-user reply relationships when aggregated into weighted edges.

Research Question 1 (Method, Findings)

The first research question of this study examines the structural organization of the Bluesky NBA reply network during the trade deadline and whether it exhibits centralized hub formation. Figure 1 shows the constructed weighted, directed reply network. In the visualization, node size corresponds to in-degree, and edge thickness corresponds to edge weight. The network contains 6,936 nodes and 5,044 edges. These counts are lower than the full dataset because isolated nodes without edges were excluded from the network.

The network exhibits a dense core surrounded by many smaller peripheral communities, producing a visual structure consistent with a core-periphery pattern. Despite this visual concentration, the network is overall quite sparse, with a density of only 0.027%. The local clustering coefficient

NBA Trade Deadline Reply Network (Directed, Weighted)

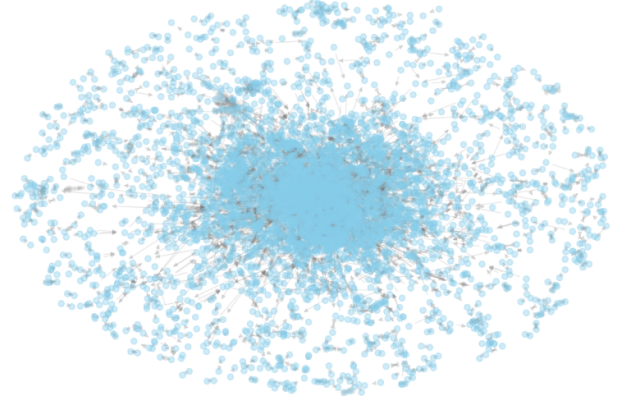


Figure 1: Bluesky NBA Reply Network during the 2026 Trade Deadline

is also low (approximately 0.031), similar to the values observed in previous studies of Twitter reply networks during exceptional events [10]. This indicates that two users replying to the same post rarely reply to each other.

The network contains 492 weakly connected components. The largest component corresponds to the central core and includes 3,688 nodes, representing 73.1% of all nodes in the network, similar to the Reddit reply networks dominated by a giant component [4]. The smaller peripheral components likely correspond to isolated reply trees in which users interact within a single conversation but do not engage with other threads. Many nodes in these components have a degree of one, indicating that they posted a single reply without further interaction. Because the giant component contains a large majority of nodes, most users have an undirected path to the core of the network.

Figure 2 and Figure 3 show that node in-degree, out-degree, in-strength, and out-strength all follow heavy-tailed, approximately power-law distributions (for a more zoomed-in view, see Figures A2-A5 in the Appendix). Most nodes have very small in-degree and out-degree values, while a small number of nodes act as highly connected hubs that receive a large number of replies. Edge weights also display a long-tailed distribution, indicating that a small number of user pairs interact repeatedly while most interactions occur only once.

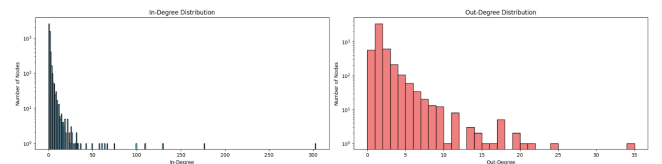


Figure 2: Degree Distribution of NBA Reply Network

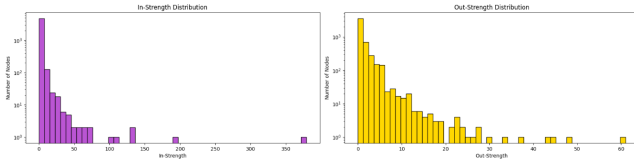


Figure 3: Strength Distribution of NBA Reply Network

Table 2 lists the users with the highest in-degree, and Table 3 shows the users with the individual posts that received the most replies. Notably, the account `shamsbot.bsky.social` has the highest in-degree in the network and appears twice among the ten posts with the highest reply counts. This account automatically reposts trade announcements from Shams Charania on Twitter and has become one of the most prominent NBA information accounts on Bluesky. Its high in-degree highlights the influence of NBA insiders and breaking news accounts in driving discussion on the platform.

Overall, information-oriented accounts dominate the high in-degree rankings. The top users include several NBA insiders and sports news organizations, indicating that accounts providing breaking news or analysis tend to attract a larger share of replies. Shamsbot also has the highest in-degree by a large margin.

Interestingly, Shamsbot is one of two accounts to appear multiple times among the ten most-replied-to posts, with Chris Hayes, a political commentator with over 800K followers. Aside from The Athletic, a sports news organization, and the Washington State Standard, most other high-reply posts come from general fans or political commentators rather than traditional sports media. This suggests that while informational accounts tend to receive a high total number of replies across posts, the most replied-to individual posts are often viral content generated by ordinary users or outside influencers.

Network centralization metrics provide further insight into the structure of the network. These metrics can be seen in Table 4. In-degree and out-degree centralization values are relatively low, indicating that no single account dominates the entire conversation, even though some accounts receive substantially more replies than others. Similarly, out-degree centralization is extremely low, suggesting that no individual user is responsible for a large proportion of replies in the network. Instead, the reply activity is broadly distributed across many participants.

Betweenness centralization is also very low, indicating that few users serve as bridges between otherwise disconnected parts of the network. This suggests that conversations tend to remain localized within individual reply threads rather than spreading through intermediary users who connect different discussion clusters.

Closeness centralization is somewhat higher than betweenness, but still relatively low overall. This indicates that while some users are structurally closer to others in the network, the network remains relatively dispersed and lacks strong global centralization.

In contrast, eigenvector centralization is substantially

Table 2: Users with the Highest Total In-Degree

Rank	Bluesky Handle	Replies	Account Type
1	<code>shamsbot.bsky.social</code>	302	NBA Insider
2	<code>theathletic.com</code>	176	Sports News Org.
3	<code>chrislhayes.bsky.social</code>	129	Political Commentator
4	<code>keithsmithnba.bsky.social</code>	109	NBA Insider
5	<code>nba.com</code>	99	Sports News Org.
6	<code>markjacob.bsky.social</code>	74	Political Commentator
7	<code>parkermolloy.com</code>	66	Political Commentator
8	<code>washingtonstatestandard.com</code>	63	News Org.
9	<code>btoporek.bsky.social</code>	60	NBA Insider
10	<code>jimowczarski.bsky.social</code>	57	NBA Insider

Table 3: Individual Posts with the Most Replies

Rank	Bluesky Handle	Replies	Account Type
1	<code>chrislhayes.bsky.social</code>	80	Political Commentator
2	<code>markjacob.bsky.social</code>	74	Political Commentator
3	<code>washingtonstatestandard.com</code>	64	News Org.
4	<code>chrislhayes.bsky.social</code>	52	Political Commentator
5	<code>shamsbot.bsky.social</code>	44	NBA Insider
6	<code>theathletic.com</code>	37	Sports News Org.
7	<code>davidbix.bsky.social</code>	32	Normal User
8	<code>lrobgunther.bsky.social</code>	32	Normal User
9	<code>parkermolloy.com</code>	31	Political Commentator
10	<code>shamsbot.bsky.social</code>	30	NBA Insider

Table 4: Freeman Centralization Metrics for the Network

Centralization Metric	Value
In-Degree Centralization	0.0596
Out-Degree Centralization	0.0065
Betweenness Centralization	0.0193
Closeness Centralization	0.0980
Eigenvector Centralization	0.4300

higher. This suggests the presence of influential hubs that are connected to other well-connected users. These hubs are likely to include NBA insiders, sports media accounts, and highly active NBA commentators who attract disproportionate attention.

Taken together, these centralization metrics indicate that conversation is largely localized within individual threads while still exhibiting a hub-dominated structure around multiple popular users. This is consistent with the earlier observation that the reply network follows a core-periphery-like pattern and exhibits power-law-like degree distributions.

The network exhibits a reciprocity of 0.359 and an assortativity of -0.096. The relatively high reciprocity indicates that replies are bidirectional roughly one-third of the time, suggesting that users often engage in short conversational exchanges. The slightly negative assortativity indicates that low-degree nodes tend to interact somewhat more frequently with degree-degree nodes. In practice, this means that highly visible accounts—such as news sources or prominent commentators—often receive replies from regular users rather than these hubs primarily interacting with other highly connected account.

Overall, the reply network is sparse and exhibits a core-periphery-like structure centered around several prominent hubs, as evidenced by the heavy-tailed degree and strength distributions. NBA insiders and information-oriented accounts consistently receive the largest number of replies, while many of the most replied-to individual posts originate from ordinary users whose content temporarily goes viral. At the same time, discourse occurs largely within localized reply threads, with relatively few users interacting across multiple conversations. These localized threads form the periphery of the network, while a small number of highly visible information sources function as hubs within the network’s core.

Research Question 2 (Method, Findings)

The second research question examines how the structure of the reply network evolves in response to major trade events. Figure 4 shows the number of NBA-related Bluesky posts per hour during the trade deadline period, overlaid with the official trade deadline and the times at which Shams Charania first reported major trades (“Woj bombs”) on Twitter.

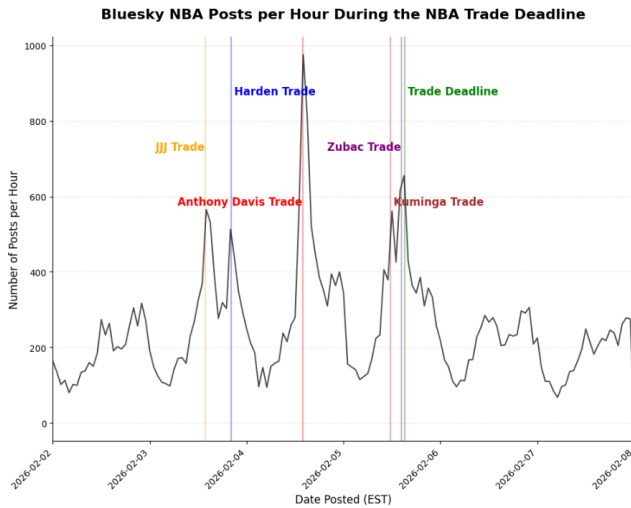


Figure 4: NBA Posts per Hour on Bluesky

Post activity exhibits a clear daily cycle, with higher activity during daytime hours and lower activity overnight. Beyond this normal temporal pattern, several pronounced spikes in posting activity are visible. Notably, all six of the largest peaks correspond either to the trade deadline itself or to the timing of major trades initially reported by Shams Charania. Most of these trades involved All-Star-level players and represented the most significant transactions of the deadline period. The only exception is the trade involving Jonathan Kuminga, which nonetheless generated substantial discussion due to his well-publicized conflict with his coach, Steve Kerr, [3] in recent seasons and his association with the Golden State Warriors, one of the league’s most widely followed teams [6].

These patterns suggest that major trade announcements function as discrete attention shocks that trigger bursts of

online discussion, consistent with prior research on exceptional events in Twitter networks [10].

To evaluate whether these spikes are statistically significant, Figure 5 calculates a rolling z-score for each hour of posting activity. For each point, the mean and standard deviation are computed using the previous 24 hours of data. For the earliest observations on February 2nd, where a full 24-hour window was unavailable, all preceding observations were used instead, which may produce inflated z-scores due to the smaller sample size.

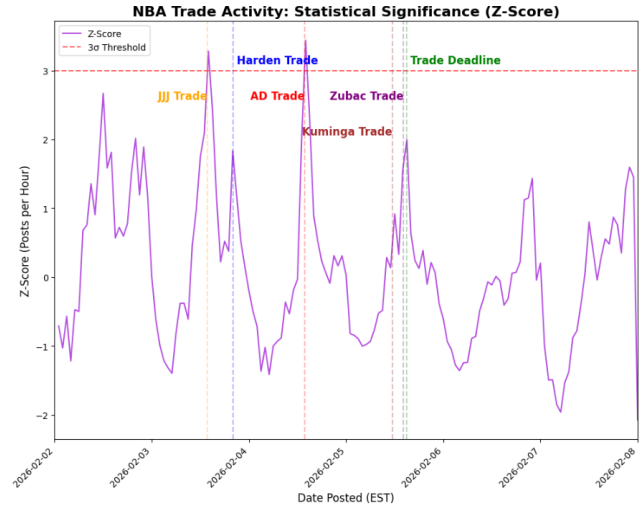


Figure 5: Rolling 24 Hour Z-Scores for NBA Posts per Hour

Using a significance threshold corresponding to a z-score of approximately 2 ($p \sim 0.05$), nearly all major trade events exceed this threshold. The elevated z-scores observed early on February 2nd are likely artifacts of the limited rolling window rather than genuine anomalies, as the baseline posting activity during that period was still stabilizing. Two events—the trades involving Jaren Jackson Jr. (JJJ) and Anthony Davis—produce z-scores above 3 ($p \sim 0.003$), indicating extremely rare deviations from normal activity. These trades involved two of the most prominent players who were moved during the deadline and produced the largest observed spikes in discourse.

Overall, these results indicate that major trade announcements, particularly those involving high-profile players, generate statistically significant surges in NBA-related posting activity on Bluesky. This supports the interpretation that trade news functions as an event-driven catalyst for bursts of online discussion, similar to the attention dynamics previously observed during exceptional events on other social media platforms [10].

Conclusion

This report analyzed the structural organization of the Bluesky NBA reply network during the 2026 NBA trade deadline using data collected through the Bluesky API. It examined whether the network exhibits centralized hub formation and how discourse activity evolves in response to major

trade events.

The reply network displays a sparse, core–periphery–like structure in which many small conversations occur on the periphery while a smaller set of highly visible accounts forms the network’s core. These central hubs largely consist of information-oriented accounts, such as NBA insiders and sports media outlets. The heavy-tailed in-degree distribution highlights the unequal concentration of attention, with a small number of users receiving a disproportionate number of replies. At the same time, the individual posts receiving the most replies were often created by ordinary users whose content temporarily went viral rather than by the largest hubs themselves.

Temporal analysis shows that NBA-related posting activity follows a clear daily cycle, decreasing during nighttime hours and increasing during the day. Superimposed on this pattern are sharp spikes in activity immediately following major trade announcements reported by Shams Charania and around the trade deadline itself. Several of these spikes are statistically significant when evaluated using a rolling 24-hour z-score, with two events exceeding a z-score of three. These results are consistent with prior research demonstrating that exceptional real-world events can generate sudden bursts of social media discourse [10].

Although this analysis focuses specifically on Bluesky discussions surrounding the NBA trade deadline, the observed structural and temporal patterns likely generalize to other social media platforms, like Twitter, given their similar interaction mechanisms and previous findings on event-driven discourse. Future work could expand the scope by incorporating additional keywords to capture a broader set of NBA-related posts, comparing trade deadline activity to baseline periods during the NBA season, or examining multiple trade deadlines across different years.

Overall, these findings suggest that “Woj bombs” function as an attention shock that rapidly mobilizes online discussion around centralized information hubs within the reply network, demonstrating how NBA-related news flows through social media discourse.

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Appendix

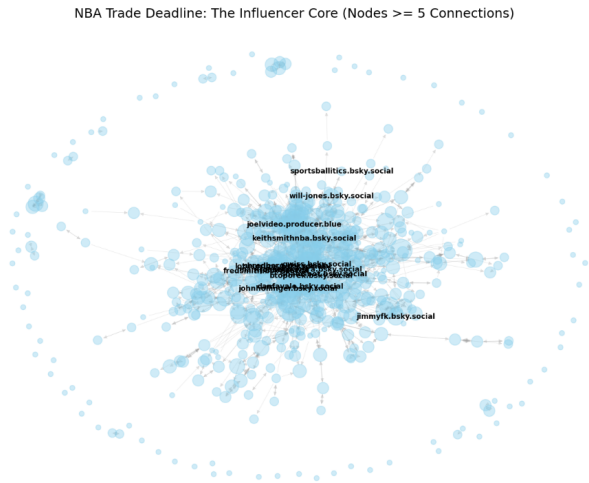


Figure A1: Bluesky NBA Reply Network Core during 2026 Trade Deadline (Degree ≥ 5 in the Original Network)

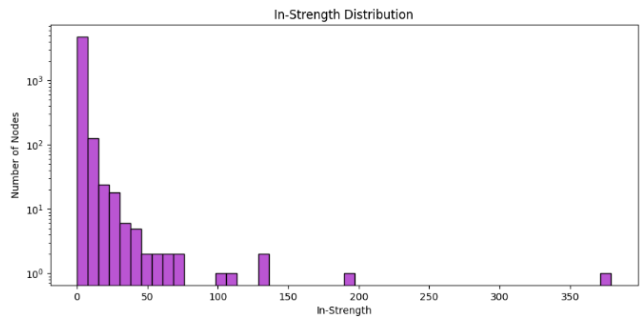


Figure A4: In-Strength Distribution of NBA Reply Network

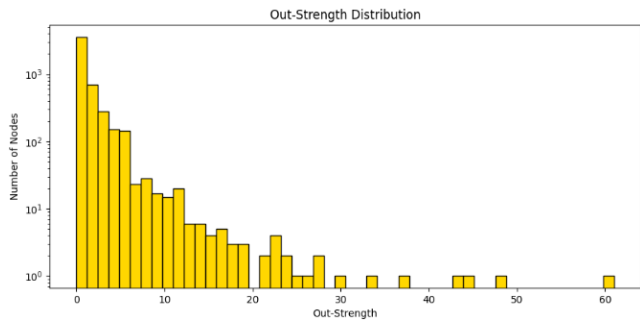


Figure A5: Out-Strength Distribution of NBA Reply Network

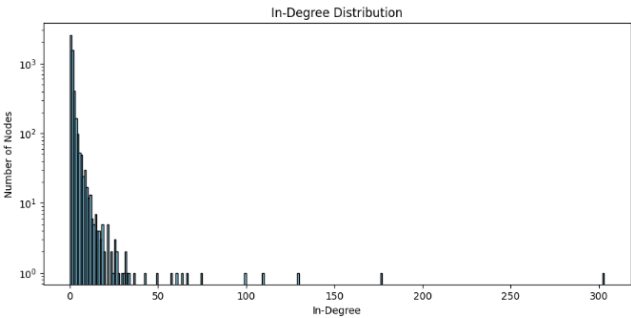


Figure A2: In-Degree Distribution of NBA Reply Network

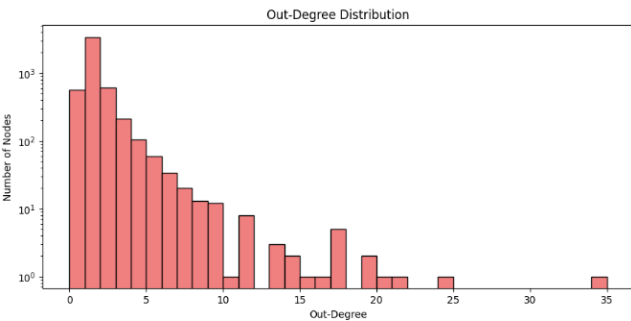


Figure A3: Out-Degree Distribution of NBA Reply Network

Table A1: Freeman Centralization Metrics for the Network Core (Degree ≥ 5 in the Original Network)

Centralization Metric	Value
In-Degree Centralization	0.1096
Out-Degree Centralization	0.0279
Betweenness Centralization	0.0981
Closeness Centralization	0.1472
Eigenvector Centralization	0.0462